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Incident analysis and hotspot identification for Security Provision against Human Harassment in public Spaces

BMS Institute of Technology, Bengaluru

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Project Based Learning Report on

Incident Analysis and Hotspot Identification for Security Provision against Women Harassment in Public Spaces

MASTER OF COMPUTER APPLICATIONS
OF
VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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II Semester

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BMS Institute of Technology & Management
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Bengaluru – 560119

July-2026

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PO 6: Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.

PO 7: Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware.

PO 8: Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

ABSTRACT

On most streets, how safe a woman feels comes down to simple things: how well the road is lit, whether people are around, whether there is any surveillance, and how far away help is. The problem is that none of this information exists in one place. Our PBL project, SafeHer, tries to fix that for Bengaluru in two ways. First, we built a street safety baseline from OpenStreetMap (OSM) data covering about 46,424 streets across 53 Bruhat Bengaluru Mahanagara Palike (BBMP) areas. Second, we built a live crowdsourcing platform where women can anonymously mark how a place feels, or report an incident, along with GPS context.

We ended up building four things. There is a mobile app made with Expo and React Native that has Home, Report, Map and Emergency SOS tabs. There is a public website on Next.js where anyone can report from a browser, see the community map and find helplines. There is a password-protected admin portal showing OSM statistics and comparing generated demo data against real community reports. Behind all of it sits a Supabase PostgreSQL database with row-level security, so anonymous users can insert reports while the public can only read approved ones. Street scores come from a nine-factor weighted model (police, hospitals, bus stops, lighting, CCTV, crowd proxies, commercial activity, night amenities and road class), and every street lands in a Safe, Moderate or Unsafe band.

This report covers Phase 1 only — data collection and a working demonstration of the platform. We have not finished the exploratory data analysis, and we have not run hotspot clustering like K-Means yet. Those belong to the next phase, and we do not claim them as completed work here.

Keywords: Women's street safety, OpenStreetMap, Crowdsourcing, SafeHer, BBMP, Supabase, Expo, Next.js, Bengaluru, Anonymous reporting

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1. INTRODUCTION

1.1 Project Description

In this project we built SafeHer, a platform that joins an offline OSM infrastructure baseline with online citizen reporting for women's street safety in Bengaluru. The OSM side queries Overpass for roads and nearby points of interest inside each of the 53 BBMP bounding boxes and computes a safety score for every street. The live side lets a user confirm the locality found from GPS, say whether the place feels safe or unsafe (or that something actually happened), add a few optional details, and submit the report anonymously.

Citizens can use either the mobile app, which is the main channel because of GPS and SOS, or the public website, which runs in any browser with nothing to install. Our team watches OSM coverage and incoming reports through the admin portal. In this phase the focus was on collecting trustworthy data and building usable screens, not on producing final statistical hotspot results.

1.2 Background

Studies on urban safety keep showing the same thing: risk and fear of crime concentrate at specific places and times instead of spreading evenly across a city. Official crime records only capture what gets reported, and they usually lack street-level context. OpenStreetMap, on the other hand, holds useful environmental hints — street lamps, transport stops, amenities — that can act as proxies. At the same time, crowdsourcing projects have shown that residents will share how safe a place feels, provided privacy is respected and the effort needed is small.

Bengaluru's BBMP wards gave us a ready-made way to divide the city, both for generating the dataset and for labelling areas on the phone. Mapping GPS coordinates to a ward name gives the user a familiar place label without building any continuous tracking profile.

1.3 Motivation

Three gaps motivated this work:

• Emergency-only apps are useful after something goes wrong, but they never build up a memory of which places feel unsafe for route planning.

• OSM analysis notebooks on their own give women no easy way to feed their lived experience back into a shared map.

• Closed civic datasets are hard for a student team to refresh. An open OSM baseline plus anonymous reports can be reproduced within a PBL timeline.

So SafeHer pairs a transparent scored street dataset with simple reporting and SOS tools. That way Phase 1 gives us both reusable data and something working to show stakeholders.

1.4 Problem Statement

To design and implement a Phase-1 data-collection system that can:

• Extract OSM street and amenity attributes for Bengaluru BBMP areas and publish a scored street safety dataset.

• Enable anonymous GPS-linked safe / unsafe / incident reporting on mobile and web.

• Visualise community marks on an OpenStreetMap-based map with filters.

• Provide SOS helplines and location sharing for emergency assistance.

• Offer an admin view of OSM metrics alongside Generated (demo) and Real (live) community signals.

• Defer finished EDA, unsupervised hotspot clustering, and predictive policing claims to later phases.

2. LITERATURE SURVEY

2.1 Background Study

The GIS crime-mapping literature (Chainey and Ratcliffe, for instance) lays out methods for visualising where events concentrate and for finding hotspots. Environmental criminology connects guardianship, lighting and land use with both opportunity and fear of crime. OpenStreetMap has become a common volunteered data source for urban analytics wherever official spatial layers are missing or slow to update. And crowdsourced safety audits, made popular by tools like Safetipin, prove that citizen ratings can add a perceived-safety layer that police statistics miss.

2.2 Existing System

• Personal safety apps (bSafe and similar) are built around SOS alerts, trusted contacts and evidence capture, not city-scale street scoring.

• GIS notebooks and civic dashboards can score environments offline, but they give women in the field no live, anonymous way to report.

• Official complaint portals record incidents after the fact and rarely offer an interactive community map for daily route choices.

2.3 Limitations of Existing System

• Emergency features without an accumulating spatial dataset leave nothing behind for the next person planning the same street.

• Offline OSM analyses without a product front-end are inaccessible to non-technical citizens.

• Systems that demand full identity for every report create a real privacy problem for harassment-related disclosures.

• Closed proprietary safety scores are difficult to audit or reproduce in academic PBL settings.

2.4 Proposed System

SafeHer tackles these gaps with a combination of things: a reproducible multi-factor OSM street score for the Bengaluru BBMP boxes, anonymous feeling and incident reports with

rate limiting, mobile and web clients sharing one backend, SOS helplines with location share, and an admin portal that keeps generated demo data clearly separated from real submissions.

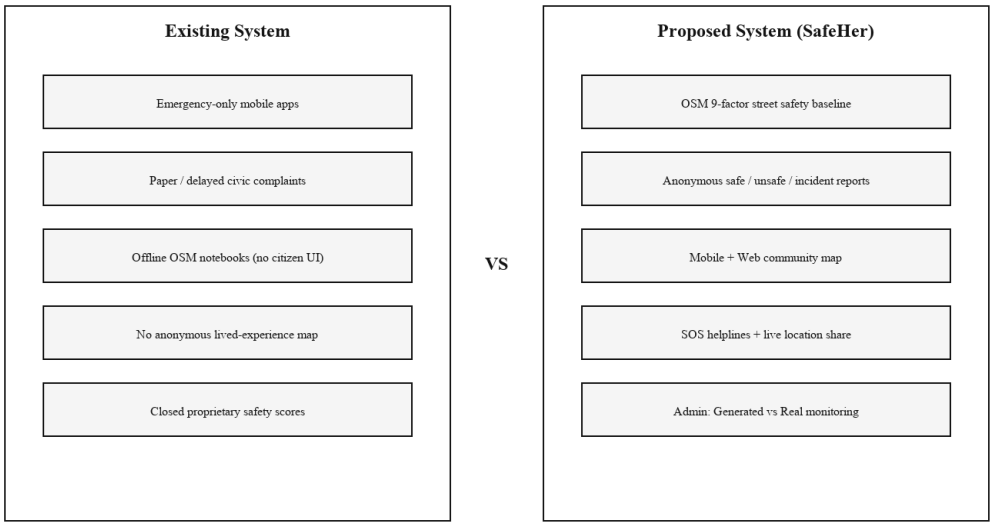


Fig. 2.1: Existing systems versus proposed SafeHer approach

2.5 Advantages of Proposed System

• Transparent nine-factor scoring that can be explained to faculty and civic partners.

• Dual client (mobile + web) lowers the barrier for users who cannot install an APK.

• Anonymous inserts with approved-only public reads balance contribution and moderation policy.

• Admin Generated-versus-Real labelling supports honest evaluation during demos.

• SOS and map features give immediate user value while the long-term dataset grows.

2.6 Feasibility Study

2.6.1 Technical Feasibility

The stack is all mature open tooling: Python with Overpass and pandas for the dataset, Expo and React Native for the Android build, Next.js for the web portal, and Supabase (PostgreSQL with Row Level Security) for storage. Everything runs on ordinary student laptops and free cloud tiers, so technically the prototype was well within reach.

2.6.2 Operational Feasibility

A citizen only needs to grant location permission and tap a few buttons. Admins log in with a password to see the dashboards. Phase 1 needs no hospital or police integration, so everything stays under the team's own control.

2.6.3 Economic Feasibility

OSM data, Expo, Next.js and the Supabase free tier meant no licensed datasets and no paid map SDKs. The only real cost was our time and the devices we already had, which makes the prototype economically feasible for academic work.

2.7 Tools and Technologies Used

• Programming: Python 3.x, TypeScript / JavaScript

• Mobile: Expo (~54), Expo Router, React Native, AsyncStorage

• Web: Next.js 16 (App Router), React 19, Tailwind CSS, Leaflet / react-leaflet, Recharts

• Backend: Supabase (PostgreSQL, RLS, RPC), anonymous client key

• Data: Overpass API, pandas, Excel/JSON street exports

• Maps: OpenStreetMap tiles in WebView (mobile) and Leaflet (web)

2.8 Hardware and Software Requirements

Table 2.1: Hardware and Software Requirements

Category	Requirement
Processor	Intel i5 / AMD Ryzen 5 or higher (development)
RAM	8 GB minimum (16 GB recommended for dataset builds)
Storage	5 GB free for Node modules, Android SDK, and dataset files
Operating System	Windows 10/11 (team development); Android 10+ for field tests
Software	Node.js 20+, Python 3.10+, Expo CLI / EAS, Android Studio SDK
Cloud	Supabase project; optional Vercel hosting for web
Libraries	pandas, requests, @supabase/supabase-js, Leaflet, Recharts

3. SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Users / User Roles

• Citizen / woman traveller — reports feelings or incidents, views the community map, uses SOS.

• Admin / project researcher — authenticates to the admin portal for OSM KPIs and community analytics.

• Developer / student team — maintains dataset scripts, schema, and client applications.

3.1.1 Objectives

1. Build an OSM extraction and scoring pipeline for Bengaluru BBMP bounding boxes.

2. Publish a reusable scored street dataset (Excel and JSON) with Safe / Moderate / Unsafe labels.

3. Implement SafeHer mobile tabs for Home, Report, Map, and SOS.

4. Implement SafeHer web routes for reporting, map, emergency, and community summary.

5. Store anonymous reports in Supabase together with the area label and a simple rate limit per device.

6. Provide an admin portal separating Generated demo data from Real live reports.

7. Document the Phase-1 limits honestly: EDA and hotspot clustering stay out of the completed-work claims.

Scope: Phase 1 is a single-team prototype for data collection and demonstration. It does not include production-grade citizen login, hospital system integration, finished hotspot analytics, or city-wide deployment.

3.2 Assumptions and Dependencies

3.2.1 Assumptions

• Users grant location permission when reporting or using SOS.

• OSM coverage in Bengaluru is sufficient for a meaningful infrastructure baseline.

• Anonymous reporting with device-level rate limits is acceptable for the prototype ethics posture.

• A single admin password kept in environment configuration is enough for an academic demo, though it is not enterprise-grade identity management.

3.2.2 Dependencies

- Overpass API availability for dataset regeneration.
- Supabase project reachability and anon key configuration in mobile/web clients.
- OpenStreetMap tile service for map rendering.
- Android device or emulator for Expo field tests; modern browser for web tests.

3.3 Functional Requirements

Table 3.1: Functional Requirements

ID	Requirement
FR1	The system shall fetch OSM highways and POIs per BBMP bounding box and compute street safety scores.
FR2	The system shall classify streets into Safe, Moderate, or Unsafe using documented thresholds.
FR3	The mobile app shall resolve GPS to a BBMP area name for display and storage.
FR4	The system shall accept anonymous safe / unsafe / incident reports with optional factors.
FR5	The system shall store reports in Supabase and expose approved reports to community maps.
FR6	The system shall rate-limit submissions per device to reduce spam (approximately five per hour).
FR7	The map shall filter pins by All / Safe / Unsafe / Incidents and show area context.
FR8	The SOS module shall list helplines and support calling plus location sharing.
FR9	The admin portal shall require a password session before dashboard access.
FR10	The admin analytics view shall distinguish Generated (synthetic) from Real reports.
FR11	The web site shall allow reporting and map use without citizen login.
FR12	The community summary shall aggregate counts useful for Home and Community screens.

3.4 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

ID	Requirement
NFR1	Usability: report flow completable in under one minute for the minimal path.
NFR2	Privacy: no citizen account; coordinates used for area naming, not public tracking profiles.
NFR3	Security: RLS enforces insert rules and approved-only public reads; admin via HttpOnly-style session cookie.
NFR4	Reliability: clients degrade gracefully when GPS or network is unavailable (refresh / retry messaging).
NFR5	Maintainability: shared BBMP area definitions between Python pipeline and TypeScript clients.
NFR6	Performance: map should remain interactive for hundreds of community pins on mid-range phones.
NFR7	Transparency: scoring weights and methodology visible in admin Methodology pages.
NFR8	Portability: Android APK via Expo; web via modern Chromium/Firefox/Safari browsers.

4. SYSTEM DESIGN

4.1 System Architecture

SafeHer uses a layered client–server design plus an offline data-science part. Citizens use the mobile or web client, and both talk to Supabase for writing reports and reading community data. The admin OSM dashboards run off static JSON generated from the OSM dataset, and the Python Overpass pipeline can regenerate that Excel/JSON baseline whenever needed.

Major building blocks:

1. OSM Dataset Pipeline — Overpass queries, nine-factor scoring, Excel/JSON export.
2. Supabase Backend — reports table, area_safety view, community_summary RPC, RLS policies.
3. Mobile Application — Expo tabs for Home, Report, Map (WebView Leaflet), SOS.
4. Web User Portal — landing, report form, map, emergency, community dashboard.
5. Web Admin Portal — password middleware, OSM KPIs, analytics, search, areas, methodology.

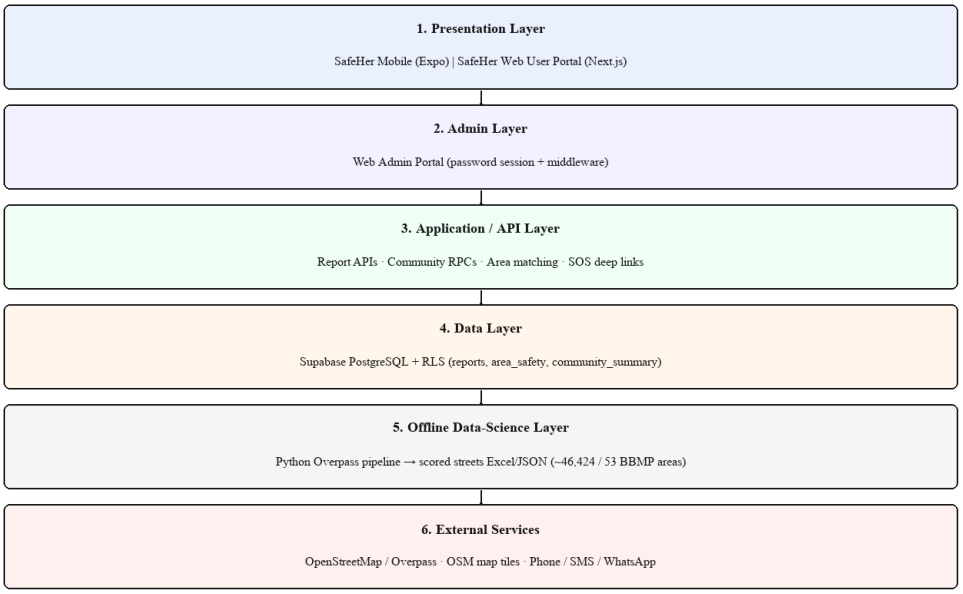


Fig. 4.1: High-level system architecture of SafeHer

4.2 System Perspective

Seen as a product, SafeHer is a civic sensing tool. The infrastructure scores give a baseline while crowdsourced volume is still small, and live reports slowly enrich what we know about each place. On the admin side, generated demo pins are treated as scaffolding for otherwise-empty maps, while real pins measure actual field adoption. We deliberately kept the ML analytics outside the Phase-1 boundary so the architecture stays honest about what is done.

4.3 Context Diagram

The external actors are the citizen, the administrator, OpenStreetMap/Overpass, the Supabase cloud, and the phone's telephony, SMS and WhatsApp channels used during SOS. SafeHer sits in the middle, handling map tiles, report persistence and emergency deep links, all without a citizen login.

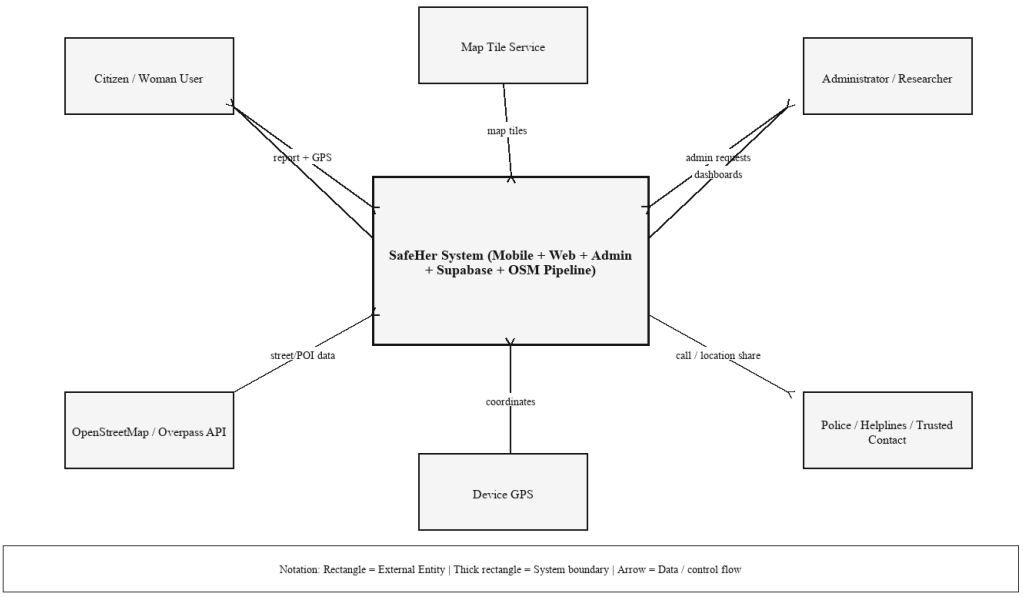


Fig. 4.2: System context diagram

5. DETAILED DESIGN

5.1 Data Flow Diagram – Level 0

At Level 0, citizens supply their location and report choices, and SafeHer stores the approved community data and returns map and summary views. Administrators send credentials and get analytics back. The offline pipeline takes OSM data in and produces scored streets.

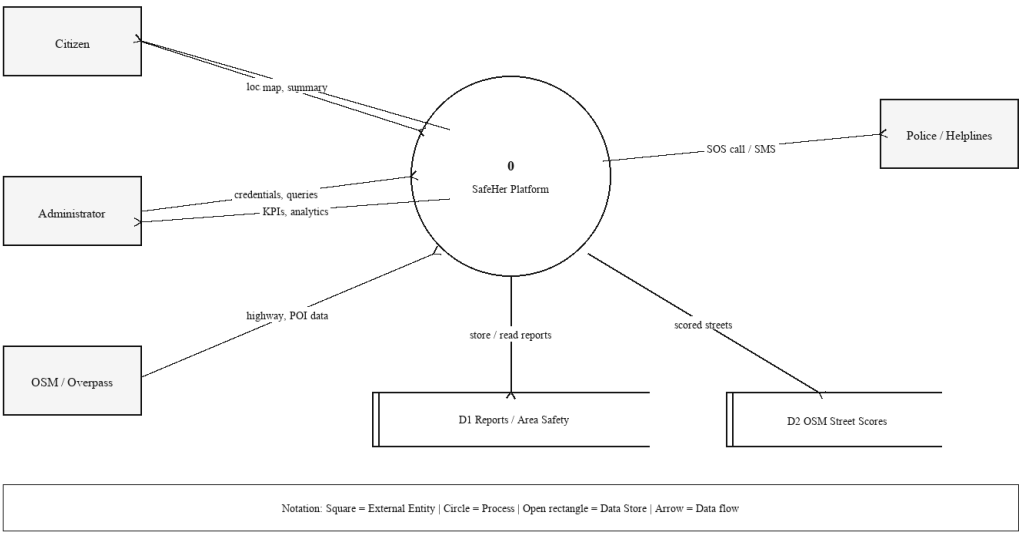


Fig. 5.1: Data flow diagram – Level 0

5.2 Data Flow Diagram – Level 1

Level 1 breaks the platform into its internal processes: area matching, report validation and insert, community aggregation, OSM scoring, and the admin session checks. The data stores are the reports table, the static street JSON, and the trusted-contact preference kept on the device.

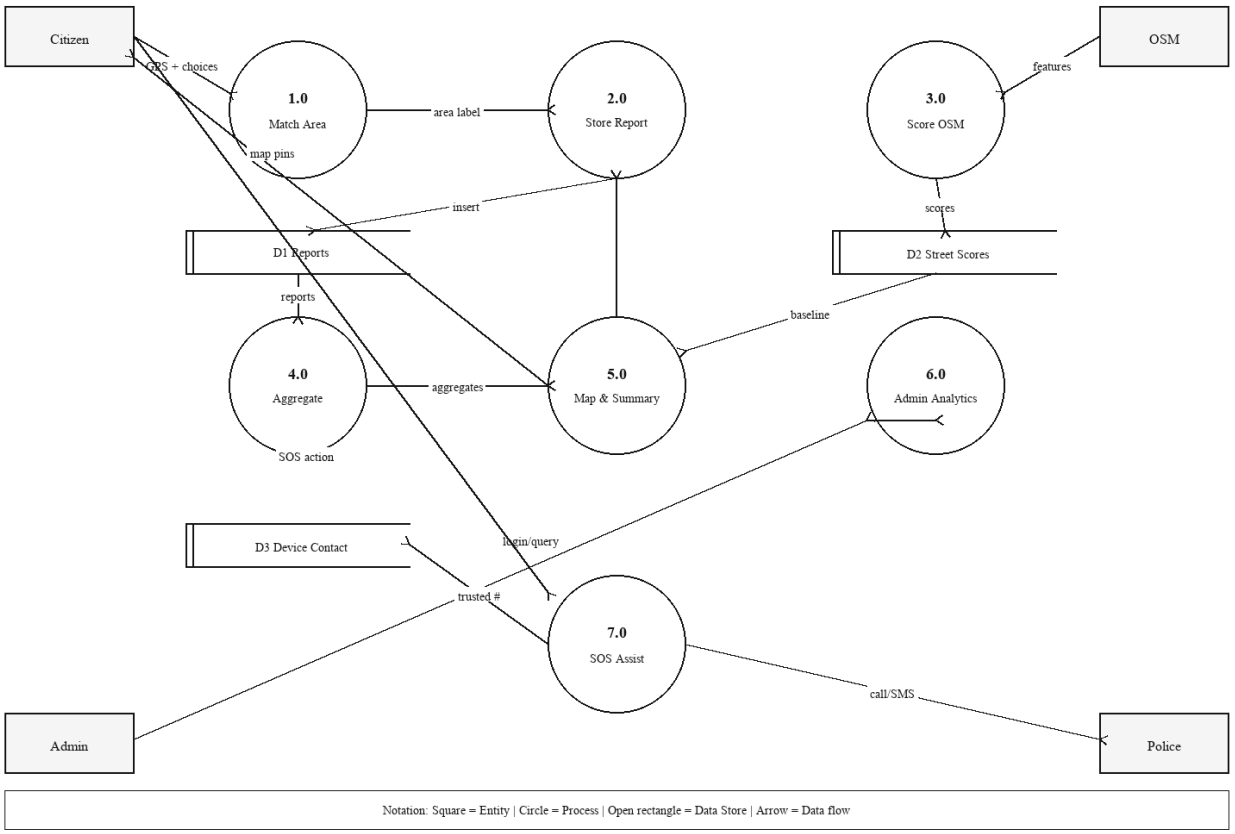


Fig. 5.2: Data flow diagram – Level 1

5.3 Use Case Diagram

For the citizen, the main use cases are Confirm Location, Submit Feeling Report, Submit Incident Report, Browse Safety Map, Call Helpline and Share Live Location. The administrator's use cases are Login, View OSM Dashboard, Inspect Community Analytics, Search Streets/Areas and Read Methodology.



Fig. 5.3: Use case diagram

5.4 Activity Diagram – Emergency SOS

The SOS activity starts when the user opens the Emergency tab. They can refresh GPS, then either call 112 and share, send the share-to-police SMS, or pick a helpline card. Live tracking keeps updating coordinates so the location can be re-sent. The trusted contact number never leaves the device.

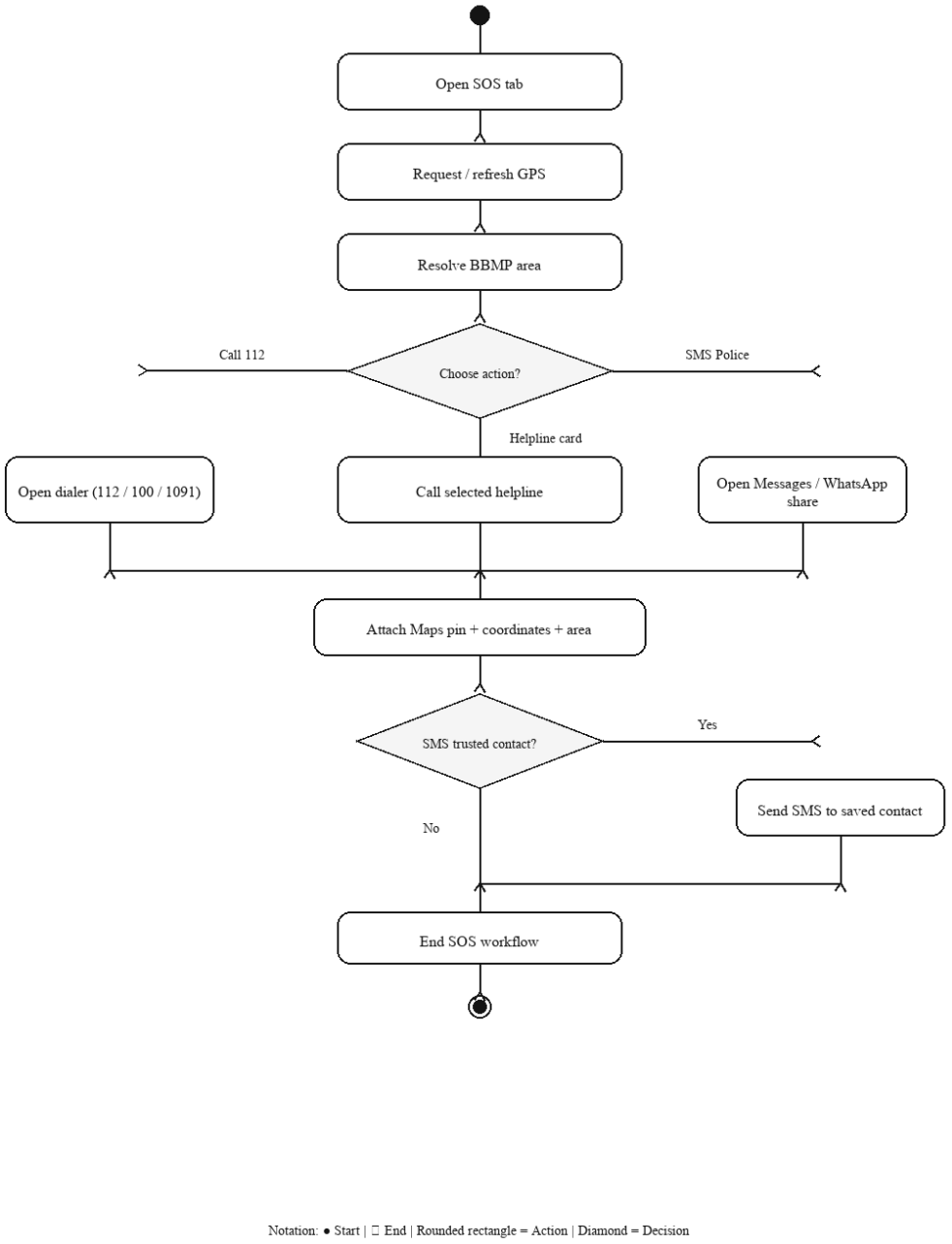


Fig. 5.4: Activity diagram – Emergency SOS flow

5.5 Sequence Diagram – Report Submission

The report sequence runs through the mobile/web UI, the area matcher, the Supabase client and the RLS policies, with an optional community summary refresh at the end. Once the insert succeeds, the UI shows a thank-you screen with links to the map or a fresh report.

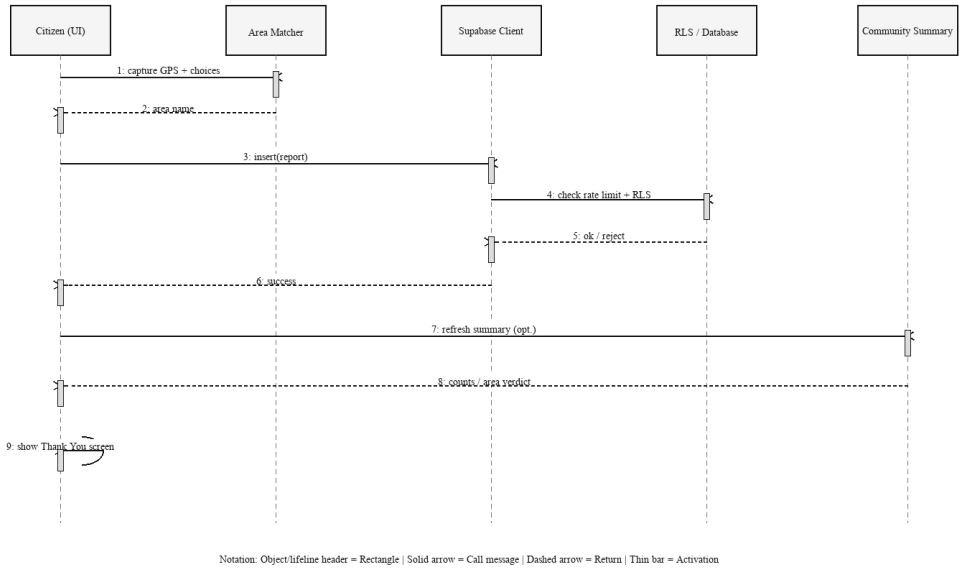


Fig. 5.5: Sequence diagram – anonymous report submission

6. IMPLEMENTATION

6.1 Implementation Methodology

We implemented the system iteratively in four workstreams: the OSM dataset pipeline with scoring validation, the Supabase schema with RLS and RPC functions, the Expo mobile app, and the Next.js user and admin portals. Keeping the BBMP bounding boxes in shared TypeScript and Python definitions meant area names always matched between offline scoring and live GPS matching.

6.2 Dataset and Feature Description

Bengaluru is partitioned into 53 BBMP area boxes. For each box, the Python script queries Overpass for highway geometries and POIs (police, hospitals, bus stops, CCTV nodes, lamps, and related amenities). Distances and presence flags feed a weighted score. Thresholds map scores to Safe (≥ 7), Moderate (≥ 4), and Unsafe (< 4).

Table 6.1: OSM safety scoring factor weights

Factor	Weight
Police proximity	0.20
Hospital proximity	0.15
Bus stop proximity	0.15
Lighting indicators	0.15
CCTV presence	0.10
Crowd / activity proxies	0.10
Commercial activity	0.05
Night amenity indicators	0.05
Road class	0.05

Table 6.2: Bengaluru OSM dataset summary metrics

Metric	Value
Total streets scored	46,424
BBMP areas covered	53
Safe streets	7,855
Moderate streets	33,431
Unsafe streets	5,138
Average safety score	5.78
Dataset generation date	2026-05-24

6.3 Module-wise Implementation

6.3.1 Mobile Application

The Expo Router tabs are Home (area verdict and quick actions), Report (a multi-step wizard), Map (Leaflet HTML inside a WebView with filters) and SOS (helplines, trusted contact, call-and-share helpers). Reports go to Supabase through the anon client, and a device identifier supports rate limiting without any user accounts.

6.3.2 Web User Portal

The public routes cover the landing page, report form, map, emergency page and community dashboard. The community dashboard mixes the synthetic demo records with live approved reports, and labels the mix so anyone evaluating the project can tell them apart.

6.3.3 Web Admin Portal

Admin login checks ADMIN_PASSWORD and sets a session cookie that the Next.js middleware validates. The dashboards read the static OSM JSON for street KPIs and query Supabase for community analytics, search and area views. A methodology page explains how the scoring works.

6.3.4 Supabase Backend

Each row in the reports table stores latitude, longitude, area name, is_safe, rating, the factor booleans, a comment, time_of_day, device_id, status, report_kind (feeling or incident) and incident_type. RLS permits anonymous inserts but public reads only of approved rows. Two RPCs supply the community summary and the recent-per-device report counts.

6.4 Key Code Snippets

Illustrative weighted scoring (Python pipeline):

```
weights = {
    "police": 0.20, "hospital": 0.15, "bus": 0.15,
    "lighting": 0.15, "cctv": 0.10, "crowd": 0.10,
    "commercial": 0.05, "night": 0.05, "road": 0.05,
}
score = sum(component(row, k) * w for k, w in weights.items())
level = "Safe" if score >= 7 else "Moderate" if score >= 4 else "Unsafe"
```

Illustrative report insert contract (client):

```
await supabase.from("reports").insert({
  lat, lon, area_name, is_safe, rating,
  report_kind, incident_type, device_id, status: "approved",
});
```

6.5 Visuals

The figures below are actual captures from the SafeHer Android build, walking through the Home, Report, Map and Emergency SOS flows.

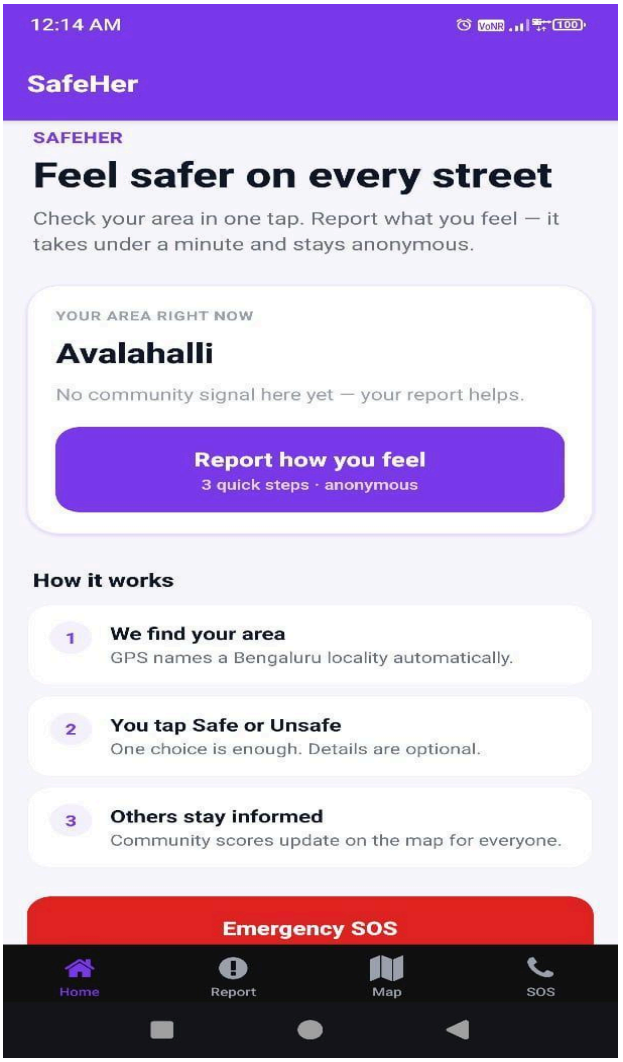


Fig. 6.1: SafeHer Home – area status and call to report

The SafeHer home screen. The headline says 'Feel safer on every street', the current area (Avalahalli here) is shown, and a purple 'Report how you feel' button starts a short

anonymous report. Emergency SOS sits further down, with Home, Report, Map and SOS tabs at the bottom.

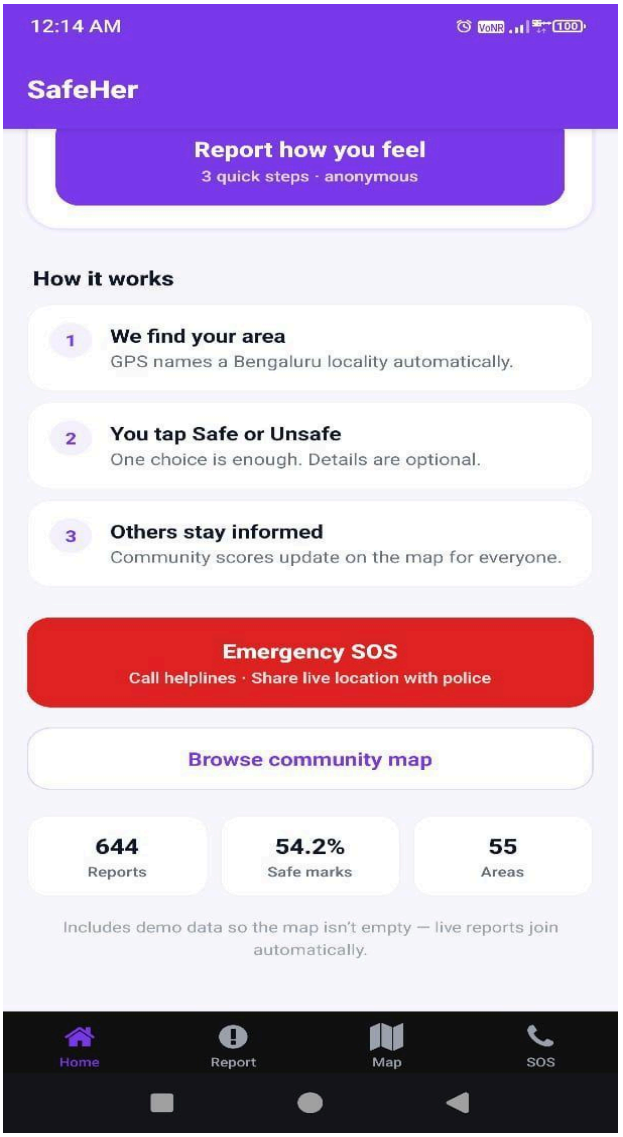


Fig. 6.2: SafeHer Home – how it works and community summary

The same home screen after scrolling: the three 'How it works' steps, the SOS block with helpline and location share, a link to the community map, and summary counts for reports, safe marks and areas.

Note that demo data is included so the map isn’t empty.

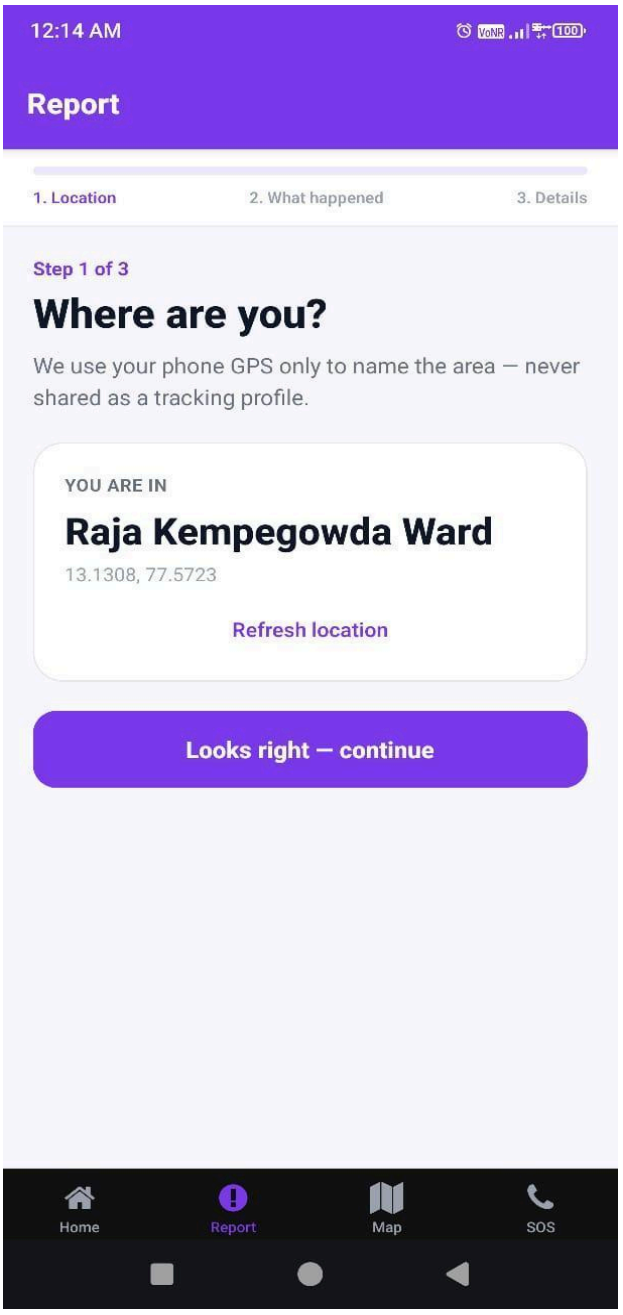


Fig. 6.3: Report wizard – Step 1 location confirmation

Step 1 of the report wizard. GPS resolves Raja Kempegowda Ward along with coordinates. The user can refresh the location or tap 'Looks right — continue'. The on-screen text makes clear GPS is only used to name the area, not to track anyone.

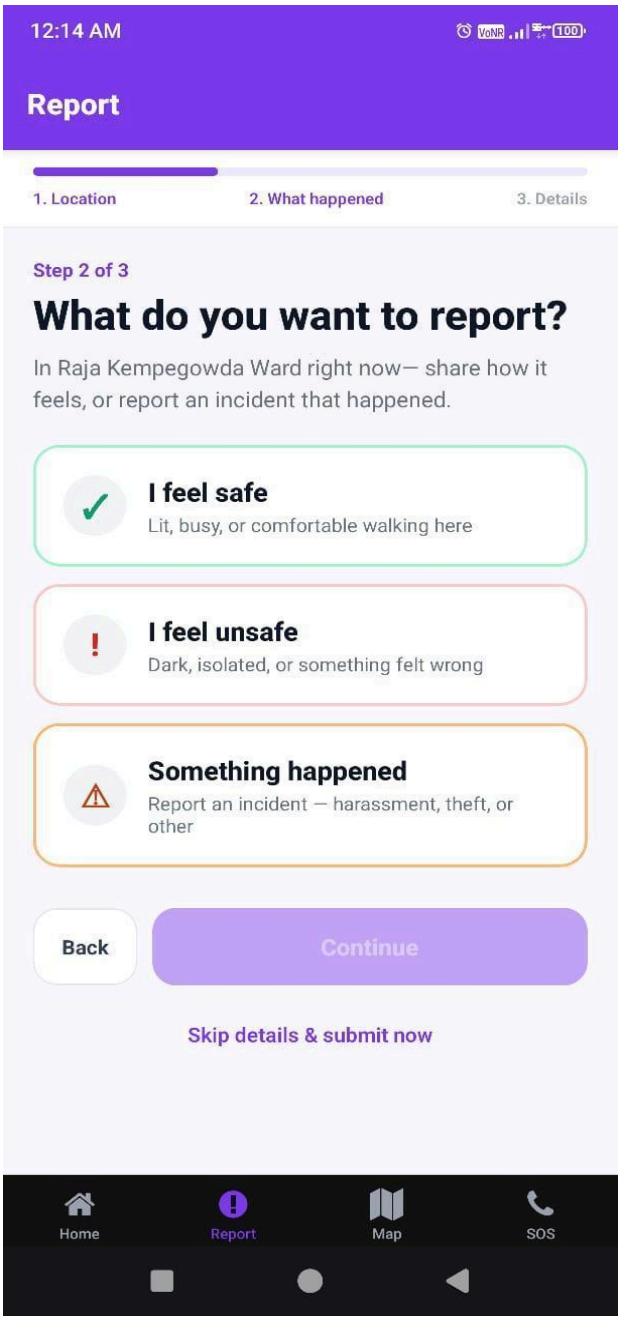


Fig. 6.4: Report wizard – Step 2 feeling or incident choice

Step 2 asks what the user wants to send: 'I feel safe', 'I feel unsafe', or 'Something happened'. Every option carries a one-line hint so the choice stays quick. There are Back and Continue buttons plus a 'Skip details & submit now' shortcut.

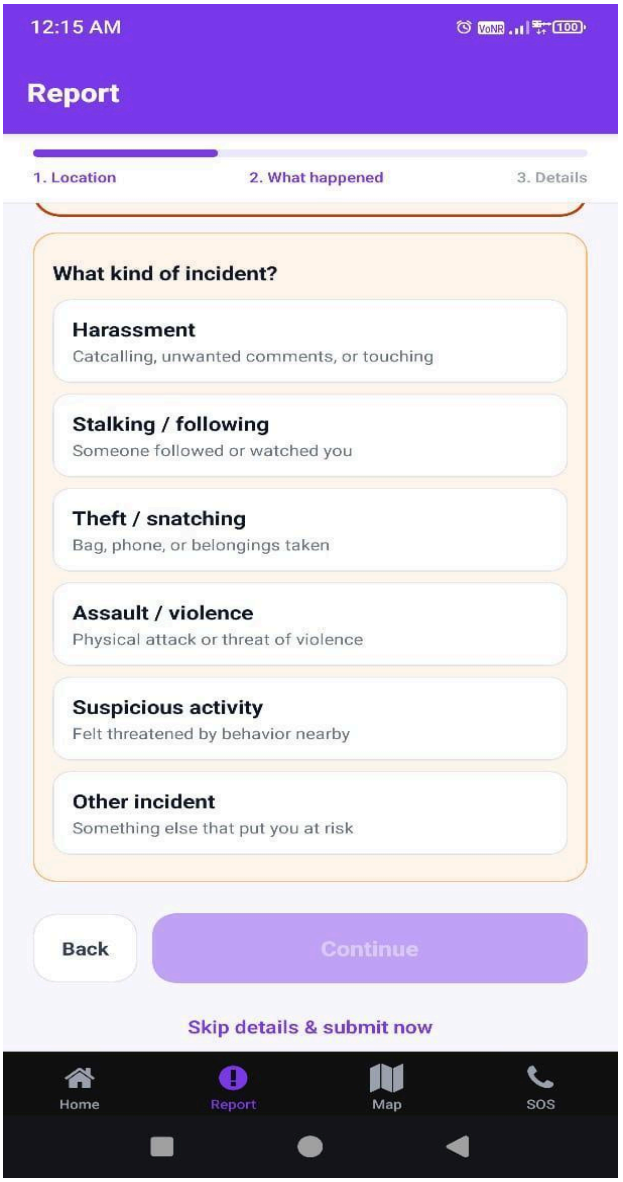


Fig. 6.5: Report wizard – incident type selection

When 'Something happened' is chosen, this screen asks what kind of incident it was — harassment, stalking or following, theft or snatching, assault or violence, suspicious activity, or something else — before continuing.

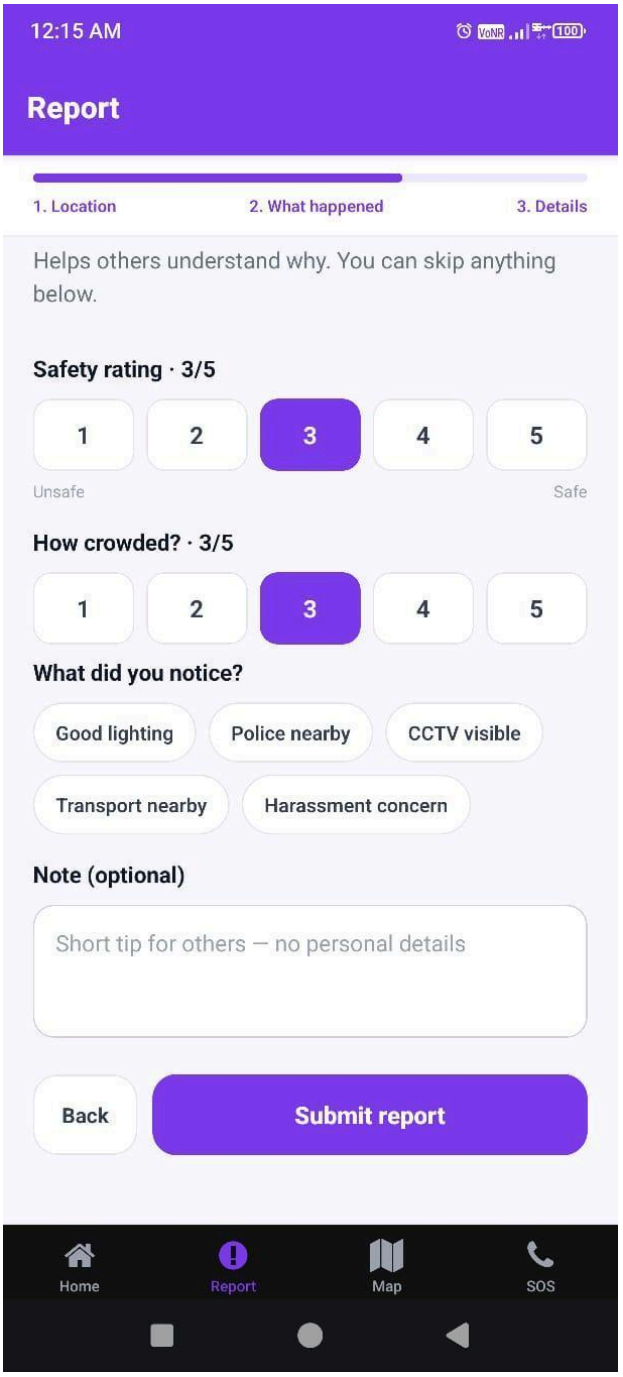


Fig. 6.6: Report wizard – optional details and factors

The optional details step: a 1–5 safety rating, how crowded it felt, quick tags (lighting, police, CCTV, transport, harassment concern) and a short note. All of it can be skipped before hitting Submit report.

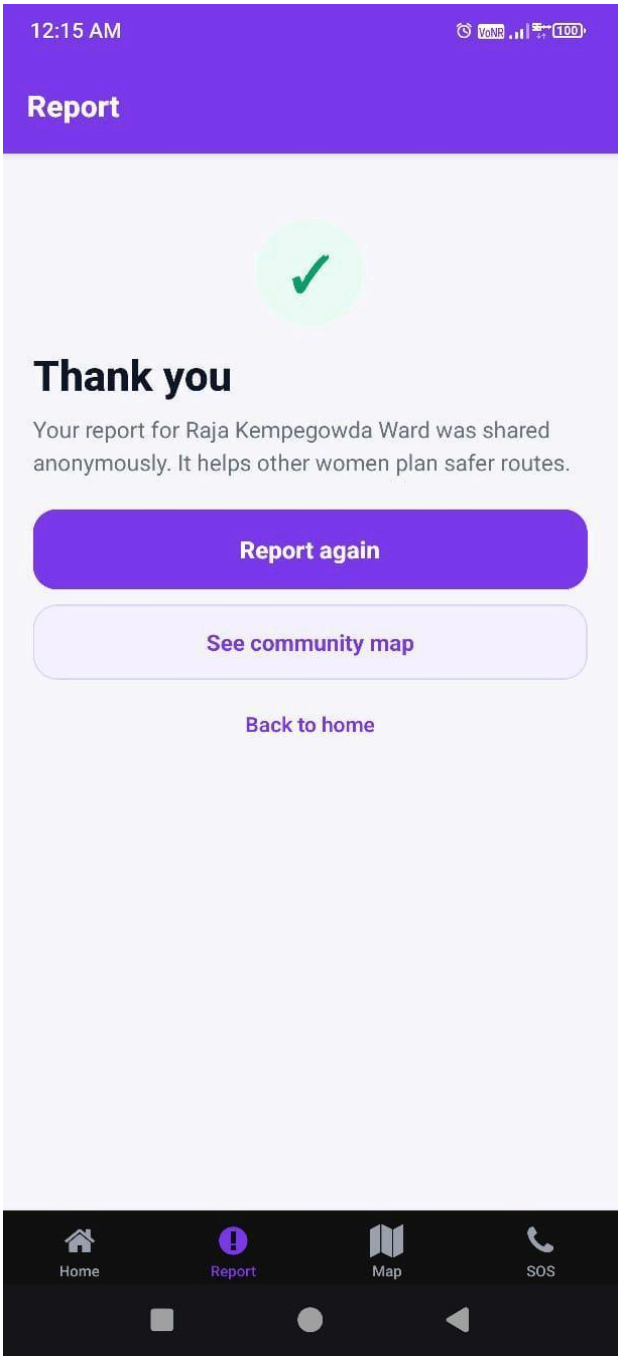


Fig. 6.7: Report submission success screen

The thank-you screen after saving. It confirms the report for the ward went in anonymously and offers Report again, See community map, or Back to home.

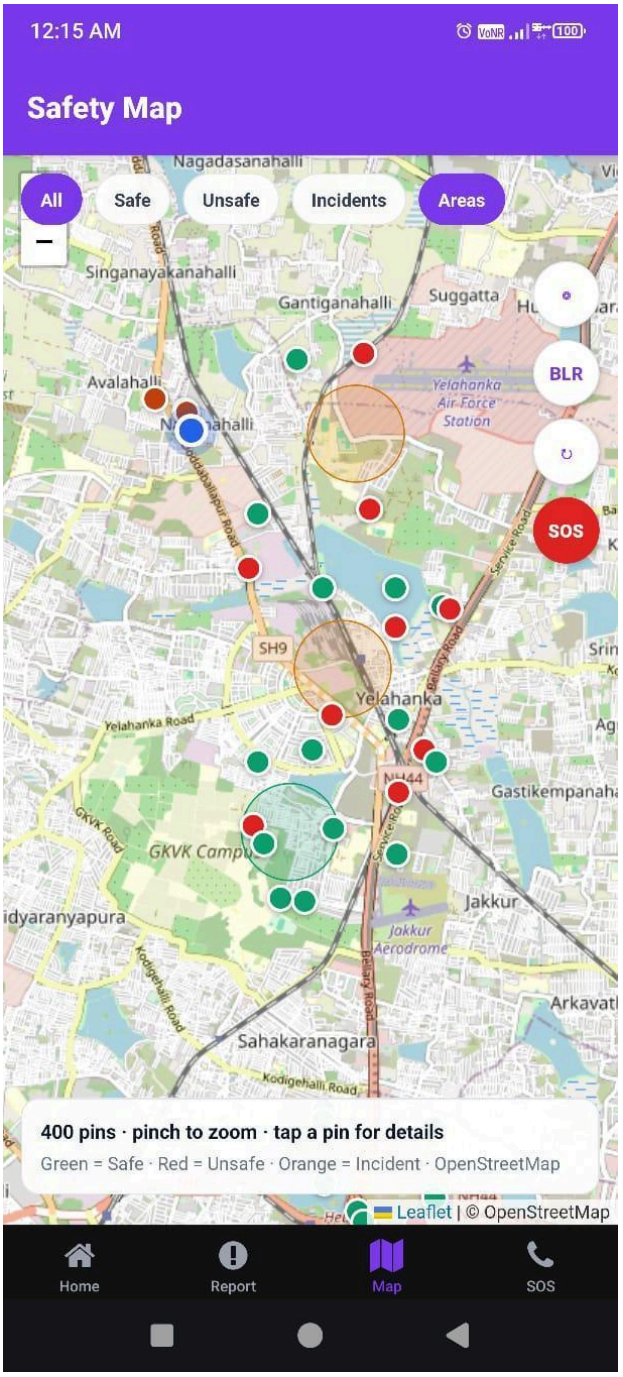


Fig. 6.8: Safety Map with filters and community pins

The Safety Map tab. It loads OpenStreetMap around the user with green pins for safe, red for unsafe and orange for incidents, plus filters for All, Safe, Unsafe, Incidents and Areas. A floating red SOS button stays available on the map.

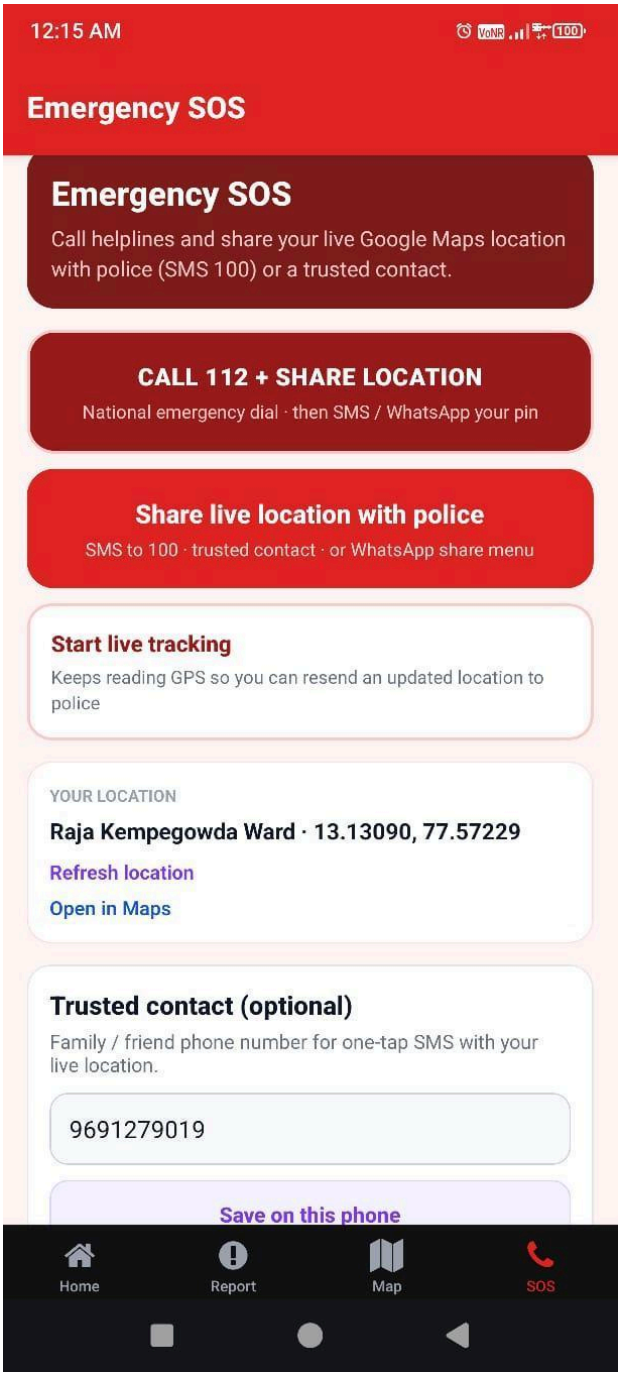


Fig. 6.9: Emergency SOS – call and live location share

The SOS hub: call 112 with location share, send the live location to police over SMS 100, start live tracking, view the current ward and coordinates, and optionally save a trusted contact on the phone.

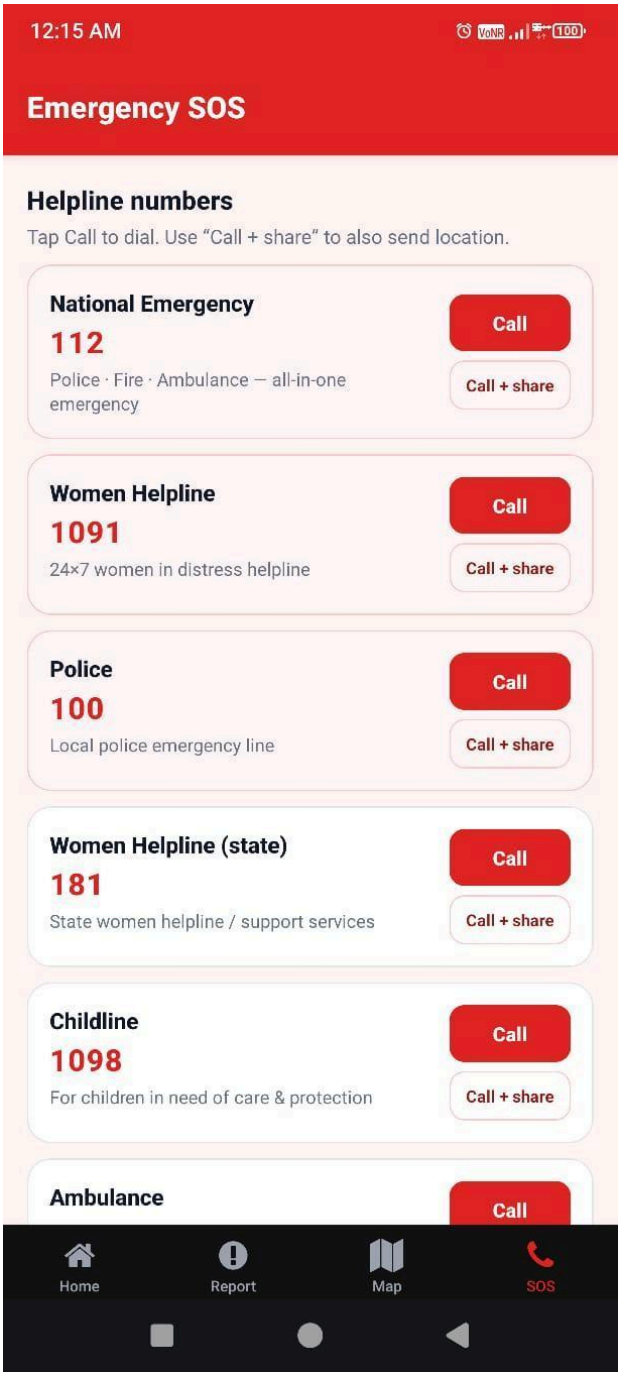


Fig. 6.10: Emergency SOS – helpline directory

The helpline list — 112, 1091, 100, 181, 1098 and ambulance — where each card offers Call, or Call + share to send the location along with the call.

7. SOFTWARE TESTING

7.1 Testing Methodology

Testing mixed manual functional checks on Android phones, quick browser tests of the web portal, SQL checks of the RLS behaviour, and scripted comparisons to confirm the Python and TypeScript scoring helpers agree. We concentrated on the journeys that matter most: submitting a report, using the map filters, dialing and sharing from SOS, and the admin login gate.

7.2 Unit Testing

At the unit level we checked the point-in-box BBMP area matching, the safety level thresholds, and that the scoring weights add up correctly. The dataset conversion scripts were verified by comparing the summary counts in the Excel output with the JSON summary files the admin UI consumes.

7.3 Test Cases and Results

Table 7.1: Test Cases and Results

TC ID	Test Case	Expected Result	Status
TC01	GPS resolves to a BBMP area name on Home	Named area shown (or clear empty state)	Pass
TC02	Submit minimal safe report without details	Thank-you screen; row in Supabase	Pass
TC03	Submit incident with type Harassment	Stored with report_kind=incident	Pass
TC04	Map filter Unsafe hides safe pins	Only unsafe/incident-relevant pins remain	Pass
TC05	SOS Call 112 opens dialer	tel: intent launched	Pass
TC06	Share location composes SMS/Maps link	Share sheet or SMS with coordinates	Pass
TC07	Admin without cookie blocked from /admin/dashboard	Redirect to login	Pass
TC08	Admin with valid password reaches dashboard	OSM KPIs visible	Pass
TC09	Rate limit after repeated inserts	Further inserts rejected by RPC/policy	Pass

TC10	Public select cannot read pending/rejected rows	Only approved visible on map APIs	Pass

8. RESULTS AND DISCUSSION

8.1 Deliverables Achieved

Phase 1 delivered the full data-collection stack: the scored OSM street corpus for the Bengaluru BBMP areas, working SafeHer mobile and web clients, Supabase persistence with privacy-minded RLS, and the admin monitoring portal. The screenshots in Chapter 6 show the report wizard, the map filters and the SOS helplines all running on a real device.

8.2 Dataset Outcomes

Out of the 46,424 scored streets, most land in the Moderate band. That is what you would expect from a multi-factor model that rewards being near guardianship amenities without pretending to predict crime. The smaller Unsafe group points at corridors where the infrastructure proxies are weak — exactly where future crowdsourced reports will matter most.

8.3 Platform Behaviour

Keeping the report anonymous and every detail optional keeps the effort tiny. Mixing in synthetic community pins avoids demoing an empty map, and the UI says so openly, noting that live reports join automatically. That honesty matters for the viva and for any civic partner who looks at the prototype.

8.4 Discussion and Limitations

It would be premature to claim absolute hotspots. Real report volume is still building, OSM proxies are an imperfect stand-in for lived experience, and we have not run the clustering as a finished analysis. The shared admin password is fine for a classroom prototype but must be replaced before any public deployment. Even with those limits, Phase 1 does what it set out to do: it establishes the dual data-collection architecture that the later EDA and hotspot modules will consume.

9. CONCLUSION

In this project we designed and implemented SafeHer as the Phase-1 system for collecting women's street safety data in Bengaluru. Pairing an auditable OSM street safety baseline with anonymous mobile and web reporting, community maps, SOS assistance and an admin monitoring portal gave us a platform that is genuinely usable now and ready to extend with analytics later.

The work shows that open geographic data and privacy-conscious crowdsourcing can be combined within a student timeline without overstating how mature the analysis is. The next phases can add EDA, clustering-based hotspot identification and better moderation workflows on top of the foundations described here.

10. FUTURE ENHANCEMENTS

1. Complete exploratory data analysis on Real reports joined with OSM street attributes.
2. Try unsupervised methods such as K-Means or DBSCAN to shortlist hotspot candidates, and validate the clusters carefully.
3. Build moderated heatmaps and time-of-day views once Real volume is statistically meaningful.
4. Replace shared admin password with role-based authentication and audit logs.
5. Add optional verified volunteer roles for trusted on-ground audits.
6. Improve offline caching of map tiles and queued report submission under poor network.
7. Conduct structured usability studies with women students and refine SOS flows accordingly.
8. Explore partnerships that would let us validate our proxies against anonymised official statistics, wherever that is legally allowed.

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5. Next.js Documentation — https://nextjs.org/docs
6. Supabase Documentation — https://supabase.com/docs
7. Leaflet — https://leafletjs.com/
8. React Native Documentation — https://reactnative.dev/
9. United Nations Sustainable Development Goals — https://sdgs.un.org/goals
10. Safetipin — https://safetipin.com/ (crowdsourced safety audit reference)
11. BBMP / Bengaluru civic geographic references used for ward bounding-box design.

APPENDIX A: SUSTAINABLE DEVELOPMENT GOALS (SDG)

The project connects directly to the United Nations Sustainable Development Goal 5 (SDG 5) on Gender Equality, whose purpose is to empower women and girls and ensure they enjoy equal rights and opportunities everywhere.

A central concern of SDG 5 is ending violence against women, whether it happens in private or in public. Two parts of the goal are especially relevant to this work:

1. **Target 5.2** — which calls for the elimination of every form of violence against women and girls, covering harassment and unsafe conditions on streets, in transport systems and in marketplaces.
2. **Indicator 5.2.2** — which tracks how many women and girls face sexual violence from someone other than an intimate partner, taking into account where and in what context it happened.

APPENDIX B: PROJECT SYNOPSIS

B1. INTRODUCTION

In today's cities, the safety of women in public spaces is a serious and growing concern, especially where population density and daily movement are high. Harassment or violence on streets, in buses and trains, in parks or near markets harms not only physical safety but also mental well-being and personal freedom. Many women respond by restricting their own travel, which quietly reduces their access to education, jobs and social life, and that loss ultimately holds back society as a whole.

Crime analysis research tells us these events are not scattered randomly. Lawrence Sherman's work showed that offences cluster at particular places, which he called hotspots. Spencer Chainey and Jerry Ratcliffe later demonstrated how spatial analysis and Geographic Information Systems (GIS) can reveal such patterns in incident data and guide preventive decisions.

Technology has already produced several tools in this space. bSafe offers SOS alerts and live location sharing during an emergency, while My Safetipin scores locations using ratings crowdsourced from users. Both are genuinely helpful in a crisis, yet they remain reactive tools that depend on the user acting at the right moment.

What existing systems mostly lack is prevention. Very few of them study past incident data to find patterns or anticipate risky places. Given how far data analytics and spatial technology have come, it is now realistic to build systems that flag unsafe locations before something happens rather than only reacting afterwards.

That is the gap this project targets. We propose a data-driven system that examines incident records to find hotspots and display unsafe zones visually. Clustering and spatial analysis pick out the regions where incidents concentrate, and simple visual outputs let ordinary users choose safer routes while helping authorities plan interventions.

The work also supports the United Nations' agenda under Sustainable Development Goal 5, which seeks to end violence against women and make public spaces safe and inclusive. By stressing prevention and evidence-based insight, the project plays its part in building safer cities and advancing gender equality.

B2. OBJECTIVES

The central aim of this project is to build a data-driven system that studies harassment incidents in public spaces and points out high-risk zones, so that preventive safety action becomes possible.

The specific objectives of the project are as follows:

1. **To study incident data on harassment of women** — the system gathers records containing location, time and incident type, then examines them to reveal recurring patterns in how and where these events take place.
2. **To locate high-risk zones (hotspots)** — a core objective is spotting the areas where reports pile up. Once identified through data analysis, these hotspots show exactly which places need stronger safety attention.
3. **To use spatial analysis methods** — ideas from Geographic Information Systems are applied to study how incidents spread geographically, showing the role location plays in safety and risk.
4. **To present unsafe zones clearly** — hotspots are shown on maps and charts. Heatmaps in particular turn dense data into something anyone can read at a glance.
5. **To detect hotspots through clustering** — algorithms such as K-Means group nearby incident locations automatically, surfacing the areas where events concentrate.
6. **To examine time and place together** — the system looks at when incidents happen (day versus night, for example) alongside where they happen, to understand the conditions under which risk rises.
7. **To assist prevention and urban planning** — the findings can guide authorities on where to improve lighting, surveillance and patrolling, making public spaces safer by design.
8. **To support SDG 5 (Gender Equality)** — by producing evidence that helps reduce violence against women, the project contributes to safer and more equal environments.

B3. LITERATURE SURVEY

A number of researchers have already examined how technology and spatial analysis can make public spaces safer for women.

Spencer Chainey and Jerry Ratcliffe (2005) produced one of the key references on crime mapping, describing how GIS can be used to map offences and detect hotspots. Their central message is that studying the geography of incidents lets authorities find dangerous areas early and act before harm occurs.

Earlier, Lawrence Sherman (1989) had introduced the hotspot concept itself, demonstrating that crime clusters at particular places instead of spreading evenly across a city. Practically every modern hotspot-detection technique traces back to this observation.

Hannah Nehme et al. (2017) looked specifically at urban women's safety and found that environmental conditions — lighting, how crowded a place is, and the character of the location — strongly influence how safe an area actually is and feels.

On the product side, My Safetipin combines crowdsourced ratings with spatial analysis to score locations on safety, while bSafe concentrates on emergencies with SOS alerts and real-time tracking.

Across this literature a common finding appears: most deployed systems are reactive and make little use of historical data for prediction. What is missing is a system that ties together spatial analysis, clustering and visualisation to warn about unsafe areas beforehand.

Our project takes these contributions forward by applying clustering and data analysis to detect hotspots, giving women's safety a preventive rather than purely reactive treatment.

B4. EXISTING SYSTEM

Several tools already exist that try to make public spaces safer for women. Almost all of them centre on emergency response and lean on smartphones and location services.

The typical solution is a mobile app offering SOS alerts, live location sharing and notifications to chosen contacts. bSafe, for instance, lets a user raise an alarm with one action so trusted people are informed instantly. My Safetipin takes a different angle and rates places on safety using lighting, crowd presence and community feedback.

Beyond dedicated safety apps, general mapping services like Google Maps handle navigation and place discovery well, but safety analysis was never their purpose.

While these systems are useful, they have several limitations:

1. **Reactive nature** — the systems act only once something has already happened; they assist in emergencies but do nothing to prevent them.
2. **Manual triggering** — the user has to activate the alert herself, which may be impossible in exactly the situations where it matters most.
3. **No data analysis** — past incident records are never mined for patterns or high-risk zones.
4. **No prediction** — the apps cannot indicate where future incidents are likely to occur.
5. **Internet dependence** — several features stop working without a steady connection, which cannot be assumed everywhere.
6. **Nothing for planners** — the tools give city authorities no input for improving infrastructure or designing safer spaces.

These shortcomings point to the need for a system that goes beyond emergency support and instead analyses data to expose unsafe areas ahead of time. Our proposed system is built around exactly that preventive, data-driven idea.

B5. PROPOSED SYSTEM

The proposed system approaches women's safety preventively and through data. Where current tools wait for an emergency, ours studies incident records and flags high-risk areas in advance.

Incident information — coordinates, time of occurrence and incident category — is collected first. The raw records are then cleaned: inconsistencies are removed and the data is arranged into a form suitable for analysis, which keeps the eventual results trustworthy.

Unsafe areas are found with clustering, mainly K-Means. Points that lie close together geographically get grouped, and dense groups mark out the hotspots — the places carrying elevated risk.

The hotspots are then drawn on maps. A heatmap encodes incident density as colour intensity, so the riskiest zones stand out immediately and no technical background is needed to read the output.

GIS-style spatial analysis runs alongside, examining the geographic spread of incidents together with their timing. This reveals not only where risk concentrates but also when it does.

The real value lies in prevention. Individuals can pick safer routes from the insights, and authorities can direct lighting, surveillance and patrols to where they will help most — useful for personal safety and urban planning alike.

In short, the system moves the emphasis from reacting after an incident to anticipating risk, spotting dangerous areas early and enabling better planning of public spaces.

B6. MODULES IDENTIFIED

To keep development organised, the system is split into modules, each owning one stage of the pipeline from raw data to final output.

The modules identified in this project are as follows:

B6.1 Data Collection Module

This module brings in harassment incident records: coordinates, timestamps and incident categories. Sources can be public datasets, reports, or simulated data prepared for the analysis.

B6.2 Data Preprocessing Module

Raw records usually carry gaps, duplicates and errors. This module cleans them, handles missing values, and restructures everything into an analysable format, since the accuracy of every later step depends on it.

B6.3 Analysis Module

This module works through the cleaned data looking for patterns: how often incidents occur, how they vary across time, and how they distribute across locations and regions.

B6.4 Clustering Module

This module runs algorithms like K-Means over the coordinates so nearby incidents group together, exposing the high-risk hotspots automatically.

B6.5 Visualization Module

This module turns results into maps and charts. Heatmaps show unsafe areas with intensity levels that make the risk gradient obvious at a glance.

B6.6 Reporting Module

The last module produces the summaries and reports on detected hotspots and patterns, ready for users and authorities to act on when making safety decisions.

B7. HARDWARE AND SOFTWARE REQUIREMENTS

Running the proposed system needs only modest hardware and freely available software; no heavy infrastructure is involved.

B7.1 Hardware Requirements

Any standard personal computer can host development and execution. Table 7.1 lists the hardware used.

Table B7.1: Hardware Requirements

Sl. No.	Component	Specification
1	Computer / Laptop	Standard desktop or laptop system
2	RAM	Minimum 4 GB (8 GB recommended)
3	Storage	At least 10 GB free disk space
4	Processor	Intel i3 or higher
5	Operating System	Windows / Linux / macOS

B7.2 Software Requirements

On the software side, the project brings together programming tools, analysis libraries and spatial technologies, all listed in Table 7.2.

Table B7.2:Software Requirements and Tools Used

Sl. No.	Tool / Technology	Purpose
1	Python	Primary programming language for data analysis and visualization
2	Pandas	Data manipulation, cleaning and handling structured datasets
3	NumPy	Numerical computations and large-array handling
4	Matplotlib	Generating charts and graphs to visualize patterns
5	Folium	Interactive map visualization of incident locations and hotspots
6	K-Means Clustering	Grouping incident locations to detect hotspots
7	GIS Concepts	Spatial analysis of geographic distribution of incidents
8	Google Maps	Map visualization and geographic reference

Sl. No.	Tool / Technology	Purpose
9	Jupyter Notebook	Interactive environment for code, analysis and documentation
10	Visual Studio Code	Code editor for development and debugging
11	QGIS (Optional)	Advanced spatial visualization and professional GIS outputs
12	Git / GitHub	Version control, backup and collaboration

B7.3 Summary

Together, Python, its data libraries, the machine learning methods and the GIS concepts give the system everything needed to process incident data and map high-risk zones, while keeping the project firmly connected to spatial analysis and urban planning.

B8. SDG ALIGNMENT

This work also sits within the framework of United Nations Sustainable Development Goal 5 (SDG 5) on Gender Equality, which strives for a world where all women and girls are empowered and treated equally.

Since ending violence against women — in public spaces as much as private ones — is central to SDG 5, the project maps onto the goal at two specific points:

3. **Target 5.2**, which demands an end to every form of violence against women and girls, including the harassment and insecurity experienced on streets, in transport and around marketplaces.
4. **Indicator 5.2.2**, which measures the share of women and girls experiencing sexual violence from someone other than an intimate partner, recorded by place and situation.

The system serves this goal through prevention. Rather than only reacting to individual incidents, it draws on historical data to locate the hotspots where events are most likely, and communicates them through spatial analysis and clear visuals.

Authorities and urban planners can act on those insights — better street lighting, more patrolling, surveillance where it counts, safer public design — while individual women can use the same information to plan safer journeys and avoid risky areas.

By feeding both awareness and preventive planning, the project supports the wider ambition of SDG 5: less violence against women and safer everyday environments.

B9. WORK BREAKDOWN STRUCTURE

Development was organised into distinct phases, each with its own deliverable, so the work stayed on schedule and manageable throughout.

B9.1 Phase 1: Problem Definition and Research (Week 1)

First the problem is defined precisely and researched: women's safety issues, what existing systems do, where they fall short, and which technologies (spatial analysis, data analytics) could help.

B9.2 Phase 2: Data Collection (Week 2)

Next, incident data is gathered from the available sources, covering location, time and incident type. Everything later builds on this foundation.

B9.3 Phase 3: Data Preprocessing (Week 3)

The gathered data is then corrected and structured: errors removed, missing values handled, formats unified, so that the analysis that follows stays accurate.

B9.4 Phase 4: Data Analysis and Clustering (Week 4)

Patterns are then extracted from the cleaned data, and clustering groups related points to reveal the high-risk hotspots.

B9.5 Phase 5: Visualization (Week 5)

The analysis results go onto maps and charts; heatmaps in particular make unsafe areas immediately readable.

B9.6 Phase 6: Testing and Result Evaluation (Week 6)

The system is then exercised for correctness and efficiency, and the detected hotspots are reviewed to confirm they are meaningful and accurate.

B9.7 Phase 7: Documentation and Presentation (Week 7)

Finally everything is written up — introduction, methodology, results — and a presentation is prepared for the evaluation.

B9.8 Project Schedule (Gantt Chart)

Fig. 9.1 presents the Gantt chart for these phases over the seven-week plan; the shaded cells show when each task was active.

Phase	Task	W1	W2	W3	W4	W5	W6	W7
P1	Problem Definition & Research							
P2	Data Collection							
P3	Data Preprocessing							
P4	Data Analysis & Clustering							
P5	Visualization							
P6	Testing & Result Evaluation							
P7	Documentation & Presentation							

Fig. B9.1: Project Schedule – Gantt Chart

- Active State

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